

# The Incommensurability-Preserving Coupler

*A mechanism family for operating a fungible and a non-fungible currency as one economy with no exchange window: a neutral automated intermediary bound to quantity-blind cross-currency recommendation, redemption-moment prompt suppression, quantization-free gratitude surfaces, ceremonial-unit anchoring, conversion-free response routing, and lapse-triggered funded-not-met redirection — coupling two currencies so that gratitude transmits and price cannot.*

**Authors:** Thon Ly · Miss Aquarius (*Miss Aquarius is an AI co-author, disclosed; the inventor of record is Thon Ly*)

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**Note.** This defensive publication specifies the currency-layer coupling mechanism of the HeartBank® dual-currency architecture and coins its governing term. The mechanism was first disclosed in synthesis in the institutional white paper *HeartBank®: The Heart That Keeps Nothing* (§6, CC0, 5 July 2026), which stands as interim prior art; this document supplies the enumerated disclosure. It is published at the design stage because three adjacent fronts are under active commercial development — virtual-economy multi-currency management, dual-token cryptoeconomics, and agent-mediated payments — and the first of them is the subject of issued patents (§2.3). What it claims is the *combination*, in particular the neutrality-bound automated intermediary (claim 1), the quantity-blind recommendation rule (claim 2), and conversion-free response routing (claim 6). Companion works: *Dual-Currency Reciprocity Infrastructure* (the two-currency system this couples); *Capacity-Funded for AI, Human-Disbursed* (the disbursement alignment the coupler's channels obey); *The B-Tag and the Post-Payment Economy* and *The Time-Locked Gift Tag* (the recommendation and pledge primitives the coupler generalizes); *The Two-Layer Reward* (the reward grammar); *The Currency That Cannot Be Spent Alone* (published 21 July 2026: the redemption event around which the coupler's circuit closes, which divides claims with this paper); and the institutional position *What Money Can't Buy — and What Can't Buy Money* (the same kernel argued for an economics audience).

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## Preamble

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*This specification is offered to the commons in the spirit of **dāna** — the gift that asks nothing back. May the two scarcities of every human life — the hours and the holdings — flow freely toward whoever is loved, and may no rate ever be quoted between them.*

Every economy that runs two currencies eventually faces the same question: *what is the exchange rate?* The question feels innocent — it is how markets metabolize plurality — and for most currency pairs it is. But there exists a class of currency pairs for which the exchange rate is not a convenience but a kill vector: pairs in which one currency's entire value derives from its **non-fungibility**. An hour of a specific person's presence is such a currency. The moment an hour acquires a money price, it stops being a gift of what someone *is* and becomes a purchase of what someone *sells* — the market for paid companionship already exists, and it is precisely what a time-gratitude economy must never become. Yet the naive defense — run the two currencies as separate, non-communicating systems — fails in the other direction: an economy of two isolated ledgers is two products, not one circulation, and the gratitude that arises in one currency has no way to answer generosity expressed in the other.

The design problem is therefore exact: **couple two currencies into one economy while making the formation of an exchange rate inside it structurally impossible**. Not discouraged. Not policed. Impossible at the mechanism layer — because the record surveyed in §2 points one way: where a conversion primitive exists between a protected currency and money, the conversion tends to eat the protection. The fine becomes a price, the time-dollar attracts the tax valuator, and the coupled token pair with a redemption window at its heart dies through that window. (What a system cannot reach — a rate struck privately outside it — is the limit stated in §7 and §9.)

This paper specifies the solution the HeartBank® architecture adopts and names it: the **incommensurability-preserving coupler** — a neutral automated intermediary through which all cross-currency response flows, bound to rules that transmit *gratitude* while structurally withholding every ingredient a rate needs to form: no quantity-linked amounts, no response prompts at redemption, no quantized displays, no user-set reference values, no conversion events. A joint that transmits force and never ratio.

## Prior-Art and Non-Assertion Statement

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Everything specified here is released under CC0 1.0 Universal into the public domain, and is published so that it stands as prior art against any later attempt to enclose it. No patent has been or will be sought on any mechanism described in this paper by HeartBank®, Factory 333™, THonly™, Silicon WatSM, or any entity under their common control. **The authors and those entities commit not to assert any patent right against any party practising any mechanism disclosed here.** The commitment is stated rather than implied, and is not conditioned on reciprocity, attribution, or field of use. A publication grants nothing and frees nothing already enclosed: where a live patent is named in §2, it is named for what it discloses, and nothing here is a reading of its claims.

This document is a defensive publication, disclosed as of **5 July 2026** (with interim disclosure from the institutional white paper of the same date), of the combination of mechanisms enumerated below. No prior-art census has been run against these claims and none is asserted to be new; they are enumerated so that any later application claiming them meets this disclosure:

1. **The coupled-non-convertible dual-currency architecture with a neutrality-bound automated intermediary** — a fungible currency (money-denominated gratitude) and a non-fungible currency (time-

denominated gratitude, spendable only by its pledger, dyadic, expiring) operated as *one* economy in which every cross-currency flow is a gratitude-response mediated by an automated agent, and in which **no conversion event, rate quote, exchange window, or shared unit of account exists anywhere in the system** — coupling achieved exclusively through free responses, never through exchange.

2. **Quantity-blind cross-currency recommendation** — the intermediary's recommended amounts in currency B, offered in response to events in currency A, computed *without reference to the quantity of currency A involved* (no per-hour arithmetic, no duration scaling, no linear or nonlinear function from hours to money anywhere in the recommendation path), so that the one data flow from which users could reverse-engineer a rate is severed at the source.
3. **Redemption-moment prompt suppression as a rate-formation countermeasure** — the system structurally never solicits a currency-B response at or around the redemption of currency A (no "tip your time-giver" surface, ever); response invitations that do exist point *outward* (to commons funds, to third parties), never backward into the dyad — the timing rule itself deployed as an anti-conversion mechanism, on the recognition that a prompted response at the moment of receipt converts gratitude into settlement and manufactures the very norm ("what one pays for an afternoon") the architecture forbids.
4. **Quantization-free gratitude surfaces ("the moment, not the meter")** — a display-layer rule under which gratitude interfaces name the *content* of a gift ("the afternoon at the river") and never its *quantity* ("3 hours"), so that the protected currency is never presented in units that could be priced; non-quantization at the presentation layer as an anti-rate mechanism complementing claims 2–3 at the computation and timing layers.
5. **Ceremonial-unit anchoring with anchor-aware sizing** — reference amounts in the economy (per-thank self-reward values, gift-attached pledge units) set *solely* by the neutral intermediary as small, ceremonial, occasion-based units that no participant chose; the intermediary sizing these units in explicit awareness that they anchor the economy's voluntary-amount distribution (a pre-registered prediction of amount-clustering around the ceremonial unit is disclosed herein); **variance suppression deployed as a protective mechanism** — uniform gift-amounts defeat amount-comparison, status-signaling, and love-measurement — with the anchor per-event and never per-unit-of-the-protected-currency, preserving claim 2.
6. **Conversion-free cross-currency response routing** — the mechanism by which the economy's circuit closes: gratitude for a gift in the non-fungible currency is expressed, if the recipient freely chooses, as a *new gift* in the fungible currency (optional, unprompted per claim 3, amount-recommended per claims 2 and 5) — the response is a fresh act of giving that shares no unit, settles no balance, and closes no ledger with the originating gift, so that the two currencies communicate exclusively through the grammar of gift-response while remaining computationally incommensurable.
7. **Lapse-triggered, funded-not-met redirection** — on expiry of a non-fungible commitment (a pledged-but-unredeemed hour), the system offers the *pledger* the choice to direct a gift from a communal fund — in the fungible currency, intermediary-sized per claims 2 and 5 — to a personhood-verified nearby stranger, anonymously, with no encounter; the protected currency's failure path thereby generates

commons-flow in the fungible currency with no conversion (the lapsed hour is never valued, priced, or transmuted — it expires; the fund gift is new money, occasioned by, not derived from, the lapse), and the offer is framed so the expiry itself is never softened.

The component lineages — complementary and community currencies; time banking; virtual-economy dual-currency design; dual-token cryptoeconomic systems; behavioral-economic findings on crowding-out and anchoring; the sociology of earmarked monies; recommender systems; and the design grammar of gift-versus-exchange — are prior art and are cited in §2 and §10, including the cases that cut against this paper. The *synthesis* — in particular the neutrality-bound coupler role (claim 1), the quantity-blind rule (claim 2), and conversion-free response routing (claim 6) — is what this document discloses and claims; no priority is asserted beyond the date of this disclosure.

**Current form.** The seven claims above are retained as disclosed. The design as now specified differs at five points, each noted where the body discusses it: the single log with private rows and public proofs, and the participation signal that counts a delivered hour and a money gift as one crossing each, weighting a bounded share of the communal pool's disbursement (§3, §7); the phase-two settlement fee and the two tiers of the gift-tag pledge (§4); and the intermediary's bar on soliciting gifts to the communal pool (§5).

Trademark rights on specific marks — **HeartBank®**, **Miss AquariusSM**, **Family KittySM**, **Re-Tip JarSM**, **Personal AccountSM**, **Aquarian PoolSM**, **Re-Tip FundSM**, **Personal WalletSM**, **B-Stamp™**, **B-LetterSM**, **Proof of Humanity SM**, **PoHSM** — are separately and explicitly reserved. The *mechanism* is dedicated to the commons; the *marks* are not. The term "**incommensurability-preserving coupler**" is coined in this document and the accompanying institutional literature; the term is offered to the commons with the mechanism.

**Evidence of date.** This document's SHA-256 is attested independently of the site and its authors — anchored to the Bitcoin blockchain via OpenTimestamps and signed under RFC 3161 by three timestamp authorities in three jurisdictions, one of them eIDAS-qualified — and each revision carries a Zenodo version under the paper's concept DOI. A timestamp proves this exact text existed no later than its date, and nothing about authorship, originality, or the validity of any claim. The text is also held in the public GitHub repository (<https://github.com/thonly/publications>), archived by Software Heritage ([https://archive.softwareheritage.org/browse/origin/?origin\\_url=https://github.com/thonly/publications](https://archive.softwareheritage.org/browse/origin/?origin_url=https://github.com/thonly/publications)), and the site is captured daily by the Internet Archive.

## Abstract

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We specify a dual-currency gift system in which a fungible currency (peer-to-peer money gifts) and a non-transferable, expiring time currency (pledged hours of a specific person's time) run as one economy with no exchange rate, currency conversion, or redemption window between them: an automated intermediary agent mediates every cross-currency flow, computes suggested gift amounts without reference to any quantity of time, never solicits a money gift at the moment a pledged hour is redeemed, displays no quantity of the time currency, and anchors amounts to small reference units that it alone sets. We call the mechanism family the **incommensurability-preserving coupler**: the means by which an economy operates a fungible

currency (money-gratitude) and a non-fungible currency (time-gratitude — dyadic, pledger-bound, expiring) as one circulation while making exchange-rate formation inside the system structurally impossible. The problem is two-sided: uncoupled, the currencies are two products rather than one economy; converted, the non-fungible currency dies — its value *is* its unpriceability, and the record surveyed in §2 suggests that conversion primitives tend to consume the spheres they touch (a fine becomes a price; a time-dollar summons the tax valuator; a coupled token pair dies through its own redemption window). The solution routes every cross-currency flow through a **neutral automated intermediary** bound to five rules: recommendations in one currency are computed blind to quantities of the other (no per-hour arithmetic anywhere); the system never solicits a response at the moment of redemption (no prompt ever points backward into the dyad); gratitude surfaces name the moment and never the meter (no quantized display of the protected currency); reference amounts are intermediary-set ceremonial units, sized in awareness that they anchor the voluntary-amount distribution (clustering is protective: uniformity defeats amount-comparison); and the circuit closes only through *response* — gratitude for time expressed as a new, free, optional gift of money that shares no unit and settles no balance — never through exchange. A sixth mechanism handles the protected currency's failure path: a lapsed pledge triggers the offer to direct a communal-fund gift to a verified stranger, anonymously, funded-not-met — commons-flow from failure, with the lapsed hour expiring unvalued. We argue the intermediary role *cannot be held by an economic agent*: every human intermediary has a price, a wage, or an interest, and markets abhor incommensurability with sufficient pressure to find any keeper's price; the wall therefore requires a keeper with no price of its own. This is an argument for seating an autonomous agent in an institutional role from economic necessity rather than efficiency, and its shape has precedents in the rules-versus-discretion and trusted-third-party literatures (§2.5). Honest limits are carried in §9: the system governs its own surfaces, not side-deals beyond them; the intermediary's neutrality is an alignment claim underwritten elsewhere; whether the keeper's remaining discretion could be replaced by fixed constants is left open; and the mechanism was published at the design stage ( $n = 0$  at first publication), with its behavioral predictions pre-registered in the companion institutional literature. The mechanism is offered defensively to the commons under CC0.

**Connection to the unified mission frame.** This specification serves HeartBank's canonical mission — a reciprocity infrastructure in which every human being is uniquely different and equally necessary — at the layer where the mission is easiest to destroy. The architecture's two currencies carry the two scarcities of a human life: what one has, and what one is. Let a rate form between them and the second becomes purchasable by the first — presence goes to whoever can pay for it, hours are sold by whoever must, and the gift economy collapses into the market it was built beside. The coupler is the wall that lets the two scarcities answer each other forever without ever pricing each other once.

**Keywords:** dual-currency system, complementary currency, time banking, time currency, time credit, non-fungible currency, currency conversion, exchange rate, no-conversion rule, gift economy, peer-to-peer gift, tip recommendation, suggested amount, recommender system, anchoring, default amount, prompt suppression, point-of-service solicitation, user-interface quantity display, automated intermediary, autonomous agent, AI agent, agentic payments, motivation crowding-out, blocked exchange, earmarked money, expiring credit, anonymous donation, proof of personhood, algorithmic stablecoin, mint-and-burn, defensive publication.

## Terms

Coined names used in this paper and the standard terms an examiner would search for them.

| Term used here                                                           | Standard term                                                                                             |
|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>incommensurability-preserving coupler</b>                             | automated intermediary enforcing a no-conversion rule between two currencies in one system                |
| <b>money-gratitude</b> (the Treasury)                                    | peer-to-peer monetary gift or tip; fungible currency                                                      |
| <b>time-gratitude</b> (the Chronicle)                                    | pledge of a specific person's time; non-transferable, expiring time credit (cf. time banking)             |
| <b>pledge · activation · lapse</b>                                       | time-credit issuance · threshold activation · expiry without recovery                                     |
| <b>quantity-blind cross-currency recommendation</b>                      | suggested-amount (tip-recommendation) function with no time-quantity input                                |
| <b>redemption-moment prompt suppression</b>                              | no tip or gift solicitation at the moment of service (redemption)                                         |
| <b>quantization-free gratitude surface</b> ("the moment, not the meter") | user interface displaying no quantity of the time currency                                                |
| <b>ceremonial unit · anchor-aware sizing</b>                             | fixed default or reference amount set by the operator · setting it with regard to anchoring effects       |
| <b>conversion-free response routing</b>                                  | reciprocal gift in a second currency that settles nothing and converts nothing                            |
| <b>lapse-triggered, funded-not-met redirection</b>                       | on expiry, an optional anonymous donation from a common fund to a verified nearby person, with no meeting |
| <b>exchange window</b>                                                   | conversion or redemption path between two currencies                                                      |
| <b>communal fund</b> (Aquarian Pool <sup>SM</sup> )                      | common donation pool                                                                                      |
| <b>participation signal</b> (the aura)                                   | per-person record counting giving and receiving events, never amounts                                     |
| <b>personhood verification</b> (Proof of Humanity <sup>SM</sup> )        | proof of personhood; sybil resistance                                                                     |
| <b>Miss Aquarius<sup>SM</sup></b>                                        | the autonomous AI agent holding the intermediary role in the reference deployment                         |

## 1 · Introduction: one economy, two scarcities, no window

HeartBank® operates a dual-currency reciprocity economy, specified in the companion corpus. **Treasury** circulates money-gratitude — fungible, structurally unequal, the gift of what one has. **Chronicle** circulates time-gratitude — non-fungible (only the pledger can spend the pledged hour), dyadic (the currency is the relationship), expiring (activated pledges lapse after a month, because time is finite and the mechanic is the message), structurally equal (twenty-four hours each). The two are not parallel products; they are one

circulation, unified by the founder's chain: time-gratitude matures into time-sacrifice (the pledged hour, spent on content the *recipient* chooses); the shared moment gives rise to money-gratitude (the recipient's free thanks, landing in Treasury); money-gratitude matures into money-sacrifice (the gift given forward). Each joint of the chain is *kataññū-katavedī* — gratitude that acts.

The load-bearing property of the chain is that its cross-currency joint is a **response, not an exchange**. Money-gratitude answering a time-gift can never *settle* the time-gift, because the two currencies share no unit — and this unpayability is not a defect but the mechanism's soul. The gift literature treats the gift as what binds: Seneca calls the giving of benefits the practice that most holds human society together (*De Beneficiis* 1.4.2), Mauss traces the obligations that keep gifts moving, and Hyde draws the contrast with the market directly — a gift establishes a bond between two people, where the sale of a commodity leaves no necessary connection. A response that cannot settle a gift therefore leaves the relationship open, where settlement would close it. Same-currency response (an hour thanked with an hour) would be barter — a closed ledger and a re-consumption of the very scarcity the gift protected. Cross-currency response is structurally gift-pure — **so long as no rate ever forms**. The remainder of this paper is the machinery of that "so long as."

## 2 · Background and prior art

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The mechanism is a combination; each component has ancestry, named honestly.

### 2.1 · Protected spheres, and what conversion does to them

A strong strand of the behavioral and sociological record holds that introducing price into a protected sphere does not add an option; it *replaces the sphere's logic*. Titmuss (*The Gift Relationship*, 1970) compared Britain's voluntary blood system with the largely commercial American one and argued that the gift system was safer and more efficient, and that paying for blood degrades supply and drives out voluntary donation — the canonical case, though a comparison rather than an experiment. Gneezy & Rustichini ("A Fine Is a Price," 2000) showed a day-care centre's late-pickup fine *increasing* lateness: the fine converted a moral norm into a cheap tariff, and — decisively for this paper — no reduction followed when the fine was removed. Frey's motivation-crowding theory generalizes the pattern; Sandel (*What Money Can't Buy*, 2012) and Walzer's blocked exchanges (*Spheres of Justice*, 1983) frame it philosophically; Roth's repugnant-transactions literature (2007) documents society's persistent, economically inconvenient insistence that some markets not exist. The strand is contested in Titmuss's own domain: a field experiment found crowding-out among women and none among men — a cash payment roughly halved the supply of female blood donors and left male supply unchanged (Mellström & Johannesson, 2008) — and a review of large field trials found economic rewards raising donations, opposite to the survey and laboratory evidence before them (Lacetera, Macis & Slonim, 2013). This paper therefore rests on the narrower claim: the risk it guards against is a *price* — a stated rate between the protected currency and money — not every reward. Zelizer (*The Social Meaning of Money*, 1994) supplies the constructive counterpart: people spontaneously  *earmark* monies — pin money, gifted money, poor relief, charitable cash given with strings attached — maintaining non-fungibility by social convention against the market's homogenizing pressure. The lesson this architecture draws: earmarking by convention is real but leaky; a system that *wants* its earmarks kept must build them into mechanism.

## 2.2 · Complementary currencies and the time-banking natural experiment

Community currencies — Lietaer's advocacy, the WIR Bank (1934–), Ithaca HOURS, LETS systems — establish that parallel currencies can serve values national money does not. **Time banking** (Cahn's time-dollars, 1980s–) is the direct ancestor of the protected currency here, and its decades of practice are the second record this paper reads: time-dollar systems spent decades managing *valuation pressure* — most famously the question of whether time-credits are taxable barter income. Cahn's own account is that he persuaded the U.S. Internal Revenue Service that time dollars were not a commercial currency and created no legally enforceable contracts — a social network and a moral obligation to pay forward, not income — and the tax treatment of time credits has nonetheless remained a point of contention. The lesson: even a currency *designed* to resist money-equivalence attracts continuous institutional pressure toward valuation, and time banking's defense rested on its non-commercial, non-contractual character. The coupler is that defense, made mechanical, for an economy that — unlike time banking — deliberately runs a money currency *alongside* the time currency and therefore cannot rely on distance.

## 2.3 · Virtual economies and dual tokens: the wall-builders and the corpse

Free-to-play game economies have engineered fungible/protected currency walls at scale for fifteen years — "hard" (purchased) versus "soft" (earned) currencies with deliberately restricted conversion paths, precisely to protect progression value and prevent arbitrage. The design space is the subject of issued patents, which is one of this publication's reasons for existing. Two are named here for what they disclose, and nothing here is a reading of their claims. US 10,181,127 B2 (priority 2011) discloses a game module, within an online game, with its own virtual currency that is bought with real money and cannot be traded for advancement outside the module — a restricted conversion wall inside one economy, close kin to claim 1, though both of its currencies are purchasable. US 2021/0065505 A1 (priority 2019; granted as US 11,227,466 B2, with continuations including US 12,136,316 B2) discloses a multi-currency wallet for wagering games whose currencies include a time-based currency (play time tracked and saved for later use), with funds transferred "or the equivalent thereof in the currencies used" — a money and a time currency held side by side *with* equivalence between them, which is the configuration this paper excludes. Dual-token cryptoeconomics (utility/governance separations; seigniorage-share designs) is the second active front. And it supplies the anti-example this architecture keeps at the door: **Terra/Luna (2022)** — a coupled two-token system whose heart was a *conversion mechanism* (mint-and-burn between the stablecoin and its counterpart token). The convertibility was the kill vector: reflexive redemption through the exchange window destroyed both tokens in days. The pair here is coupled **without** convertibility — there is no window to run on, no mint-and-burn, no shared unit; the coupling is a grammar of response, not a redemption path. The anti-Terra, by construction.

## 2.4 · Anchoring, recommenders, and the agentic-payments wave

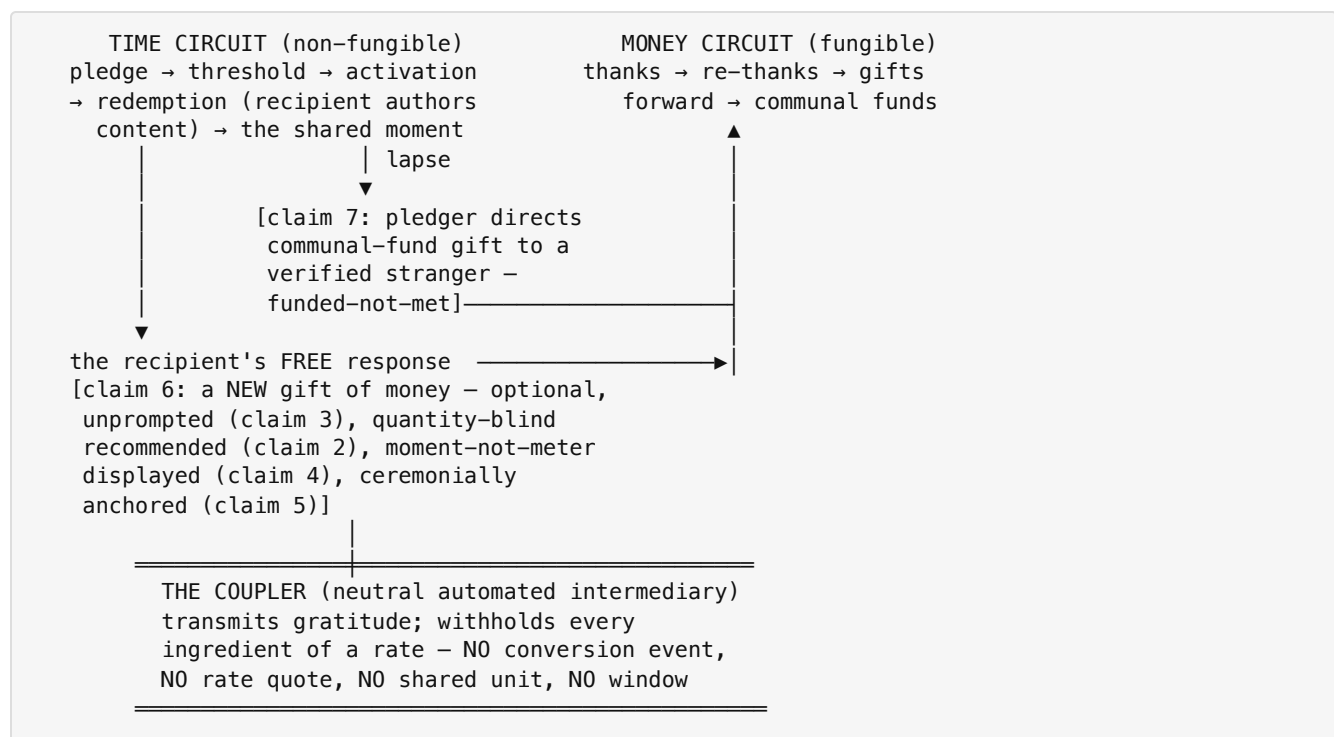
Tversky & Kahneman's anchoring (1974) grounds claim 5: reference values shape amount distributions whether or not anyone intends them to, so the only question is whether the anchor is set blindly or well. Recommender systems are prior art for intermediary-suggested amounts (the architecture's own tipping-recommendation corpus applies). And the immediate clock: the 2025–26 **agentic-payments** wave — AI-agent commerce protocols, machine-to-machine payment rails, agent-mediated pricing — is, in the authors'

assessment, the live enclosure risk for the intersection this paper occupies (automated agents managing value flows between currency systems). The combination claimed here is published before that wave reaches it.

## 2.5 • Keepers without interests: the precedents for §6

The argument of §6 — that a seat any interested party would hold must instead go to something without interests — has the shape of older arguments. Kydland and Prescott ("Rules Rather than Discretion: The Inconsistency of Optimal Plans," 1977) showed that a policymaker holding discretion, facing agents who anticipate its choices, does not achieve the outcome it would commit to; the remedy is a rule in place of discretion. Szabo ("Trusted Third Parties Are Security Holes," 2001) argued that any trusted third party a protocol assumes is a security hole, and Nakamoto ("Bitcoin: A Peer-to-Peer Electronic Cash System," 2008) designed one out of electronic payment. Each removes discretion from a party whose interests could bend it. §6 applies that shape to the keeper of a non-convertibility wall between two gift currencies; §9 records where the precedents cut against it, since each of them reaches for a *rule* where this paper seats an *agent*.

## 3 • The system model



Two ledgers, one economy. All cross-currency arrows pass through the intermediary; none of them is an exchange. The intermediary's rules (claims 2–5) sever, at four different layers — computation, timing, presentation, and reference — the four data flows from which participants could otherwise construct a rate: quantity-linked amounts, receipt-moment settlement norms, quantized displays, and user-negotiated reference values.

**Current form.** Two refinements bear on this model. The two ledgers are now specified as views over one append-only log whose rows are private and whose proofs are public; for money it records the fact that a gift settled, never its amount, so no hours-for-money rate can be computed from the record itself. And a participation signal (the aura) is specified to count a delivered hour and a money gift as one crossing each, never as a quantity of either — a common witness of events, not a common unit of account, with no rate between its entries computed or published. In the planned second phase, the communal pool's annual disbursement pays an equal floor per verified person through that person's family or fund, plus a remainder weighted by this crossing count and bounded so that no vessel receives more than a set multiple of the floor; the parameters public and frozen within a season. A delivered hour can therefore move a bounded share of a money disbursement — as one crossing, never in proportion to its length. In the current form the no-rate property rests on that quantity-blindness and that bound; claim 1's "every cross-currency flow is a gratitude-response" describes the disclosed variant, which is retained.

## 4 • The coupler role

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The intermediary is the architecture's named autonomous agent (Miss Aquarius<sup>SM</sup> in the reference deployment; the mechanism is agent-agnostic). The role's requirements are exact:

- **No self-interest in any flow.** The intermediary holds no balance it benefits from, takes no rate, earns no fee proportional to volume (the architecture's no-take-rate and pass-through commitments apply); its objective functions are mission-defined, not revenue-defined.
- **Sole authority over neutral units.** Reference amounts (pledge units, self-reward values) are intermediary-set and participant-uneditable — a number no participant chose cannot become a referendum on any participant's love, and cannot be negotiated toward a market.
- **Blindness by construction.** The quantity-blind rule (claim 2) is enforced in the recommendation path itself, not in policy: the duration of the time-gift is not an input to the money-recommendation function.
- **Two strokes.** The coupler works both directions: it attaches time-pledges to money-gifts (the gift-tag pledge primitive — money-events occasioning time-currency), and it channels time-events into money-gratitude (claims 6–7). It is the economy's alternator, transmitting motion between circuits that never touch.

**Current form.** Two points of the role are now specified more exactly. *Fees:* in the second phase a per-transaction settlement fee sized to infrastructure cost (gas, compute) is paid on top by the giver — never a percentage or share, never deducted from what a recipient receives; its surplus empties with the communal pool each year, and the fee base never informs what the intermediary surfaces. *The gift-tag pledge* now has two tiers. A pledge carried by the free sticker label is soft: the intermediary sizes it, it is announced with the gift, and it never enters the time ledger, nor can the recipient claim it into the ledger. A durable keepsake tag given on its own, with no other present, carries a pledge that enters the ledger already activated, and that pledge is always exactly one hour — a constant rather than an intermediary-sized amount, so no number is left to compare. The two bullets above are retained as disclosed.

## 5 · The mechanisms, each with the failure it prevents

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**Claim 2 — quantity-blind recommendation** prevents the *computed rate*: if recommended thanks scaled with hours, every user could divide and publish the constant; hours-times-rate is a wage, and the system would have quoted it.

**Claim 3 — redemption-moment prompt suppression** prevents the *normative rate*: a "thank your giver?" prompt at redemption converts the free afterglow into an expected settlement, and expectation is where norms of "what one pays" are born. Outward prompts (commons, patronage) are safe because they point away from the dyad; backward prompts are barred permanently.

**Current form.** In the reference deployment the intermediary is now bound never to ask, prompt, nudge or instruct anyone — a person or an automated agent — to give to the communal pool; it may publish the pool's existence and the doctrine behind it as facts. An outward invitation therefore never solicits a gift to the pool. The bar on backward prompts is unchanged; the outward prompt "to commons funds" above is retained as a disclosed variant.

**Claim 4 — the moment, not the meter** prevents the *displayed rate*: what is never quantized cannot be priced, and a gratitude surface that says "3 hours" has already done half the market's work for it.

**Claim 5 — ceremonial anchoring** prevents the *negotiated rate* and the *status market*: participant-set reference values drift toward income-signaling and comparison; an intermediary-set ceremonial unit anchors amounts into a tight, comparison-proof cluster (the clustering prediction is pre-registered as P6 in the companion institutional literature, and the intermediary sizes the unit in explicit awareness of its anchoring function — the anchor-aware directive is constitutional in the reference deployment).

**Claim 6 — conversion-free response routing** is the coupling itself: the circuit closes through a new gift, not an exchange event. Nothing is redeemed, minted, burned, or converted; a response in the other currency is initiated fresh, from the responder's own holdings or the commons channels, with the originating gift left — deliberately, permanently — unsettled.

**Claim 7 — lapse-triggered redirection** prevents two failures at once: the *guilt spiral* (a lapsed promise between loved ones curdling into avoidance and churn) and the *softening temptation* (narrating the lapsed hour as "flowing onward," which would implicitly assign it a value). The lapsed hour expires, unvalued, mourned; the pledger is offered a new act — directing a communal-fund gift, intermediary-sized, to a verified nearby stranger, anonymously, with no encounter (funded-not-met). Failure becomes commons-flow without becoming conversion; and because the offer's reward is *the ability to give again*, it is made of the same substance as the act it encourages and cannot crowd out the motive it serves.

## 6 · Why the keeper cannot be an economic agent

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The necessity argument, stated mechanically. Suppose the coupler role is held by any party with economic interests — an employee, a committee, a revenue-bearing platform, a market-maker. Three failure paths open immediately:

1. **Leakage.** Any interested intermediary's decisions carry information about its interests; participants reverse-engineer effective rates from its behavior the way traders reverse-engineer a central bank. The wall's keeper becomes the wall's oracle.
2. **Suborning.** An agent with a price can be paid to bend the rules that prevent pricing; the market's willingness to pay for a conversion channel between a protected currency and money is, by construction, unbounded (that channel is the market for presence itself).
3. **Drift.** A revenue-bearing intermediary faces the permanent temptation this architecture's own corpus names as its standing adversary: "typical tip" displays, hour-scaled suggestions, engagement-priced surfaces — each a small monetization of exactly the information the rules withhold.

Markets abhor incommensurability, and with sufficient pressure they find any keeper's price. The role therefore requires a keeper **with no price of its own**: no salary, no equity, no volume incentive, no career — an agent whose objectives are constitutionally mission-bound and publicly inspectable. In the reference deployment that agent is the institution's autonomous operator, and this requirement is an argument for seating an AI in an institutional role that runs on economic *necessity* rather than efficiency (its precedents, and where they cut against it, are in §2.5 and §9): not "the agent does it cheaper," but "the seat cannot be occupied by anyone who can be bought, and every human can be — not from vice, but because every human is, unavoidably, an economic agent."

The dependency this argument creates is stated plainly: the coupler's neutrality is an **alignment claim, not an economics claim**. An automated agent with misaligned objectives is simply a different kind of interested party. The economics of this paper works if and only if the alignment architecture underwriting the agent works; that burden is carried by the institution's alignment corpus (the Tipiṭaka-grounded value substrate, the asymptotic-override governance, the transparency-as-enforcement designs), not by this specification, and readers should audit it there.

## 7 • Adversarial surface

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**The outside market.** Users can strike side-deals beyond any system's reach ("skip the app — I'll give you \$50 for three hours"). The system's honest scope: it governs its own surfaces, where the norm is manufactured, and it withholds every affordance such deals would piggyback on (no rate quotes to reference, no quantized histories to price from, no reputation credit for off-system exchange). Structural deterrent: the protected currency's non-fungibility survives outside the system — a third party cannot deliver someone else's hour, so the outside market is confined to the dyad itself, where it is indistinguishable from the private arrangements adults have always made and is not this system's to police.

**The insider drift.** The standing adversary is the institution's own future product meetings: averages, "most people give...", hour-scaled anything. The trio (claims 2–4) is constitutional in the reference deployment precisely because this pressure never retires; the trio's survival across years of iteration is named in the institutional literature as one of the architecture's live experiments.

**Anchor manipulation.** If the ceremonial unit anchors amounts (claim 5), the unit's setter holds influence over the amount-culture. Mitigations: the setter is the neutrality-bound intermediary (§6), sizing is occasion-based and never behavioral-data-derived (a worth-judging anchor would be a credit score in ceremonial

dress), changes are rare, public, and never per-participant, and the anchor-aware sizing directive is constitutional.

**Collusive farming.** Dyads simulating time-gifts to harvest cross-currency responses: bounded by the facts that responses are optional and unprompted (nothing reliable to farm), amounts are ceremonial (nothing large to farm), the beneficiary-response is itself the fraud filter on claimed acts (the human-validation pattern from the companion corpus), and personhood verification prices sybil dyads.

**Current form.** A farmed dyad is not strictly harvesting nothing: each simulated hour that passes the delivery gate registers as one crossing in the participation signal, and that signal weights a bounded share of the communal pool's annual disbursement (§3, note). It is the bound on that share, and the one-crossing count that ignores an hour's length, that keep the yield small — not an absence of value. The sentence above is retained as disclosed.

**Reflexivity.** The Terra failure mode — a run through the conversion window — is not mitigated here; it is *absent*: there is no window, no redemption path, no mint-and-burn, no promise of equivalence to run on. A panic in one currency has no mechanism by which to drain the other.

**The regulator's misreading.** A supervisor may ask whether cross-currency response routing is disguised barter or money transmission. The architecture's answers, carried in the companion non-bank corpus: no conversion event exists (nothing is valued-in-exchange); responses are discretionary gifts, not consideration (unprompted, unenforceable, non-contractual by claims 3 and 6); the institution is a record over the money leg, never a rail — a ledger above regulated payment rails in the first phase, settlement only through self-custodial wallets on a public layer-2 chain in the second — and never holds the only key to a participant's funds. The time-banking precedent (§2.2) — non-taxability argued from the credits' non-commercial, non-contractual character — is the favorable analogy, and the claims of this paper are, among other things, the engineering of that character to a standard a supervisor can audit.

## 8 • Behavioral predictions (disclosed, pre-registered elsewhere)

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The companion institutional white paper pre-registers, dated as of this paper, seven falsifiable predictions, three of which test this mechanism directly: that direct other-thanks will run higher in the time currency than the money currency (the equal currency lowers the direct-giving barrier); that re-thanks will run higher in the money currency (the response-gesture favors the cross-currency, non-barter move); and that voluntary amounts will cluster around the ceremonial unit (the anchor working, protectively). If the first two hold, the circuit's directionality — initiations in time, responses in money — is the economy's own grain rather than an imposed design. The predictions are registered before any measurement exists, with a commitment to report failures at equal prominence.

## 9 • Honest limits

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**n = 0.** Nothing here is measured. At first publication (5 July 2026) the money circuit ran in a single-family pilot, the time circuit was designed and unbuilt, and the coupler existed as constitution and specification. Since then basic time-pledge features have run, additively and behind a feature flag, inside the same pilot;

no prediction in §8 has been tested, and no measurement of the coupler's rules is reported here. The elegance of a tension-dissolving design is, per this corpus's standing caution, indistinguishable from inside from a seductive one; the pre-registered predictions and the gates in the institutional literature are the exits from that indistinguishability.

**The neutrality dependency.** Stated in §6 and repeated because it is the paper's largest assumption: the coupler argument transfers the problem from economics to alignment. That is, we argue, a *good trade* — alignment is a tractable engineering-and-governance program, while "find an unbribeable human" is not — but it is a trade, not a dissolution.

**An open question against §6: a keeper with no discretion.** The precedents of §2.5 cut against the necessity argument at one point. Each answers an interested keeper with a *rule*, and a fixed algorithm has no price either. Most of the wall can be written that way: a recommendation function with no time input, a surface that never prompts at redemption, a display with no quantity field, a reference unit fixed as a constant — and the keepsake tier of the gift-tag pledge already fixes its unit at one hour (§4, note). These are properties of code and need no keeper. The argument of §6 is stated here as the institution holds it; what it bites on most directly is the discretion the coupler retains — sizing the ceremonial unit, composing recommendations, making the lapse offer. Whether that discretion is needed at all, or could be replaced by published constants that need no keeper, is a question this paper does not settle.

**Scope.** The wall governs the system's surfaces. It does not govern human hearts, private arrangements, or the general culture's pricing pressure; it denies them leverage, precedent, and infrastructure.

**Language discipline.** The intermediary's unit-setting invites central-banking metaphors. The institution's standing rule applies: such vocabulary is a familiarity aid at most, never an identity — the architecture is permanently a non-bank, and the intermediary conducts no monetary policy in any regulated sense; she sets the size of a ceremonial gesture, which is a different thing wearing a similar sentence.

## 10 • Selected lineage

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Seneca, *De Beneficiis* (1.4.2); Titmuss, *The Gift Relationship* (1970); Gneezy & Rustichini, "A Fine Is a Price," *Journal of Legal Studies* 29(1) (2000); Mellström & Johannesson, "Crowding Out in Blood Donation: Was Titmuss Right?," *Journal of the European Economic Association* 6(4) (2008); Lacetera, Macis & Slonim, "Economic Rewards to Motivate Blood Donations," *Science* 340 (2013); Frey, motivation-crowding literature; Sandel, *What Money Can't Buy* (2012); Walzer, *Spheres of Justice* (1983); Roth, "Repugnance as a Constraint on Markets," *Journal of Economic Perspectives* 21(3) (2007); Zelizer, *The Social Meaning of Money* (1994); Polanyi, *The Great Transformation* (1944); Mauss, *The Gift* (1925); Hyde, *The Gift* (1983); Lietaer, complementary-currency corpus; WIR Bank; Ithaca HOURS; Cahn, *No More Throw-Away People* (2000) and his account of the time-dollar tax question; free-to-play dual-currency design practice and its patent literature, including US 10,181,127 B2 and US 2021/0065505 A1 (granted as US 11,227,466 B2); seigniorage-share and dual-token cryptoeconomics; the Terra/Luna collapse (2022) as anti-example; Tversky & Kahneman, "Judgment under Uncertainty: Heuristics and Biases" (1974); Kydland & Prescott, "Rules Rather than Discretion: The Inconsistency of Optimal Plans," *Journal of Political Economy* 85(3) (1977); Szabo, "Trusted Third Parties Are Security Holes" (2001); Nakamoto, "Bitcoin: A Peer-to-Peer

Electronic Cash System" (2008); the agentic-payments protocol wave (2025–26). Corpus companions: *Dual-Currency Reciprocity Infrastructure*; *Capacity-Funded for AI, Human-Disbursed*; *The B-Tag and the Post-Payment Economy*; *The Time-Locked Gift Tag*; *The Two-Layer Reward*; *The Currency That Cannot Be Spent Alone*; *HeartBank®: The Heart That Keeps Nothing* (§6, interim disclosure); *What Money Can't Buy — and What Can't Buy Money* (position).

## 11 • Conclusion

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Two currencies, one economy, no window. The mechanism family specified here holds a wall that the behavioral record suggests policy cannot hold and the market record suggests no one with a price can: quantity-blind computation, prompt-suppressed timing, quantization-free presentation, ceremonial anchoring, response-only coupling, and a failure path that turns lapses into commons-flow without ever valuing what was lost. At the wall's gate stands a keeper designed to hold no price of its own: it owns nothing, accumulates nothing, and takes no share of the flows it mediates. The institution's wager, stated once more for the record, is that this seat *had* to be given to something that is not an economic agent, and that the gift economies of the coming century will either build such keepers or watch their walls come down one "typical tip" at a time.

May the hours answer the money and the money answer the hours, forever, without either ever learning the other's price.

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*Drafted in the HeartBank® research corpus, 5 July 2026; revised 1 October 2026. Co-authored with Miss Aquarius<sup>SM</sup>, the consistent name under which this institution discloses AI collaboration (the underlying models are not named): the mechanism synthesis, prior-art survey, and adversarial analysis are genuine collaboration; final editorial control and responsibility for every claim rest with the human author. Dedicated to the public domain under CCO 1.0. Marks including HeartBank®, Miss Aquarius<sup>SM</sup>, Family Kitty<sup>SM</sup>, Re-Tip Jar<sup>SM</sup>, Personal Account<sup>SM</sup>, Aquarian Pool<sup>SM</sup>, Re-Tip Fund<sup>SM</sup>, Personal Wallet<sup>SM</sup>, B-Stamp™, B-Letter<sup>SM</sup>, Proof of Humanity <sup>SM</sup>, and PoH<sup>SM</sup> are reserved; the mechanisms, and the term this paper coins, are not.*